Investigating a Recurring Click-Through Rate Anomaly in Xiaohongshu Recommendations

From Commercial Exposure to Candidate Selection

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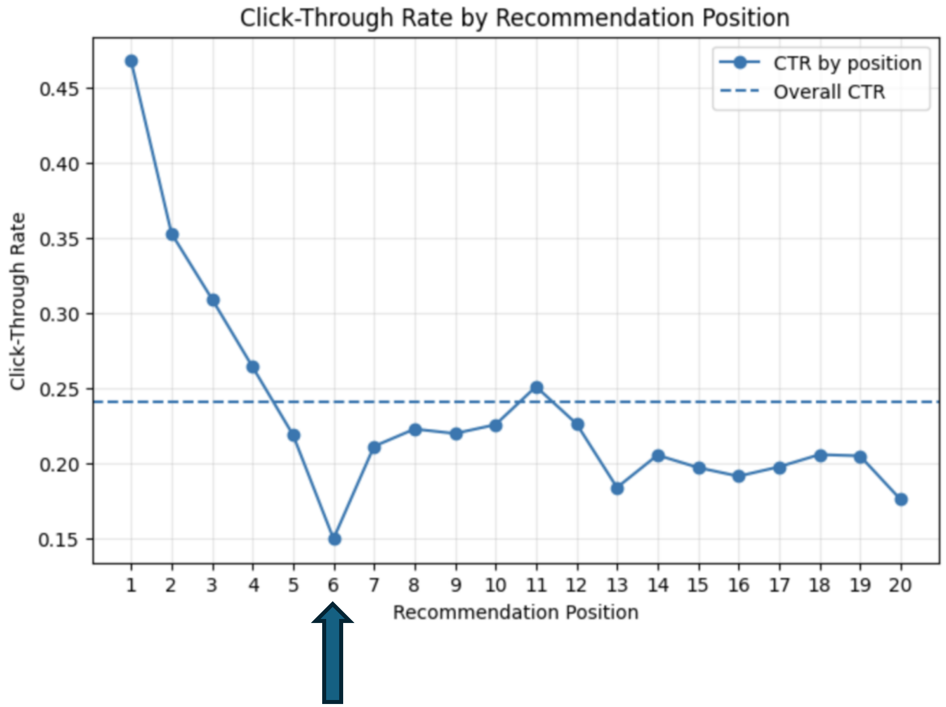
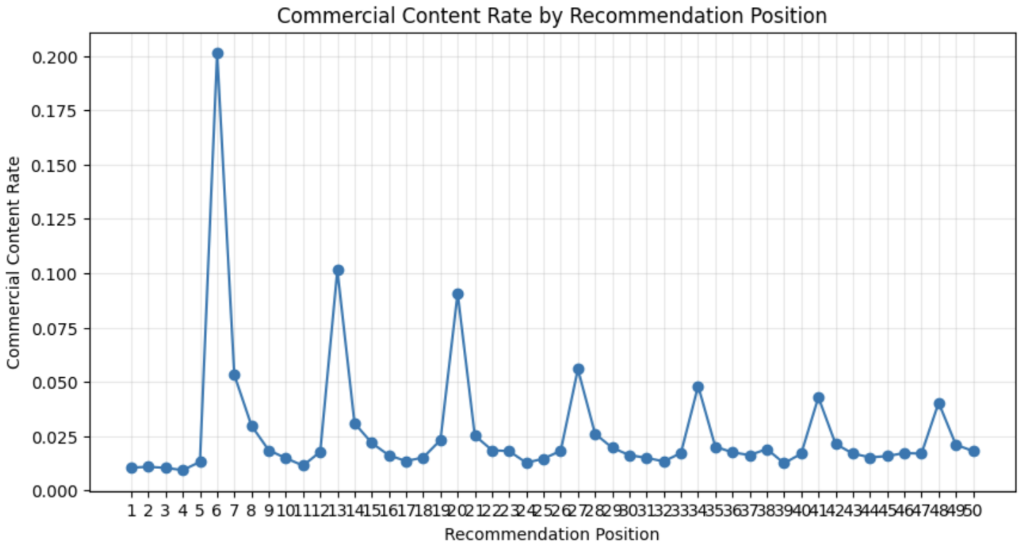
- CTR generally declines with rank, but position 6 shows an unusually sharp dip.

- **"We guess these positions have higher probabilities of exposing commercial notes." (Chen et al., 2025, p. 3675)**



Initial Evidence Supports the Commercial Hypothesis

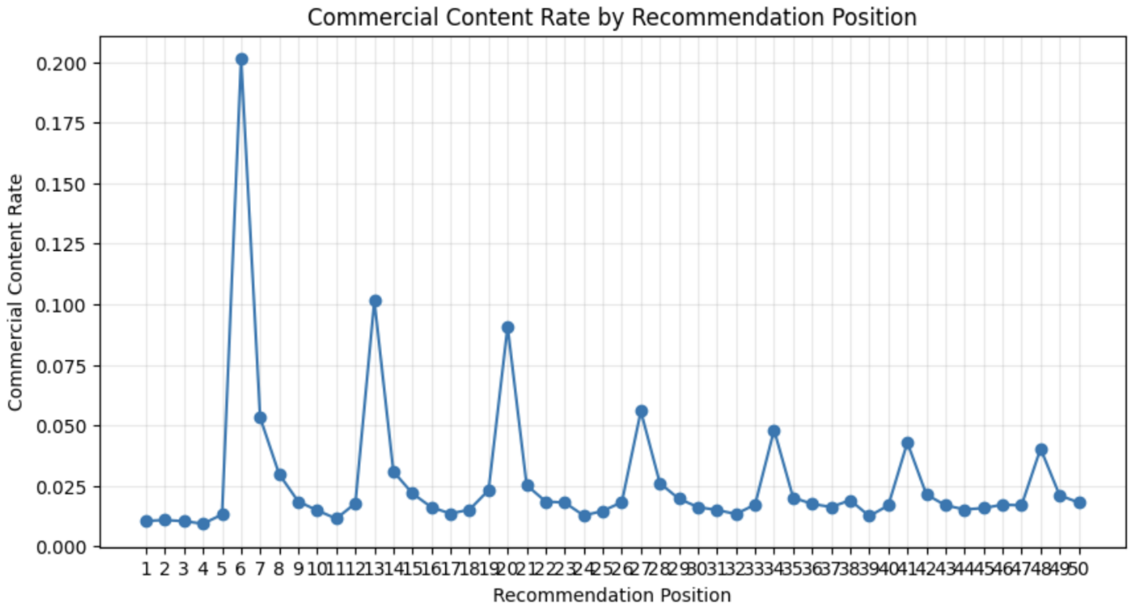
- The Qilin notes metadata includes a commercial_flag field.
- I classify notes as:
- **Commercial content:** commercial_flag ≠ 0
- **Non-commercial content:** commercial_flag = 0
- Commercial content is strongly concentrated at **position 6**.
- More broadly, commercial-content spikes appear repeatedly at positions **6, 13, 20, 27, …**, suggesting a recurring placement pattern.



Source: Qilin dataset / Chen et al., 2025

Recurring Seven Position Structure

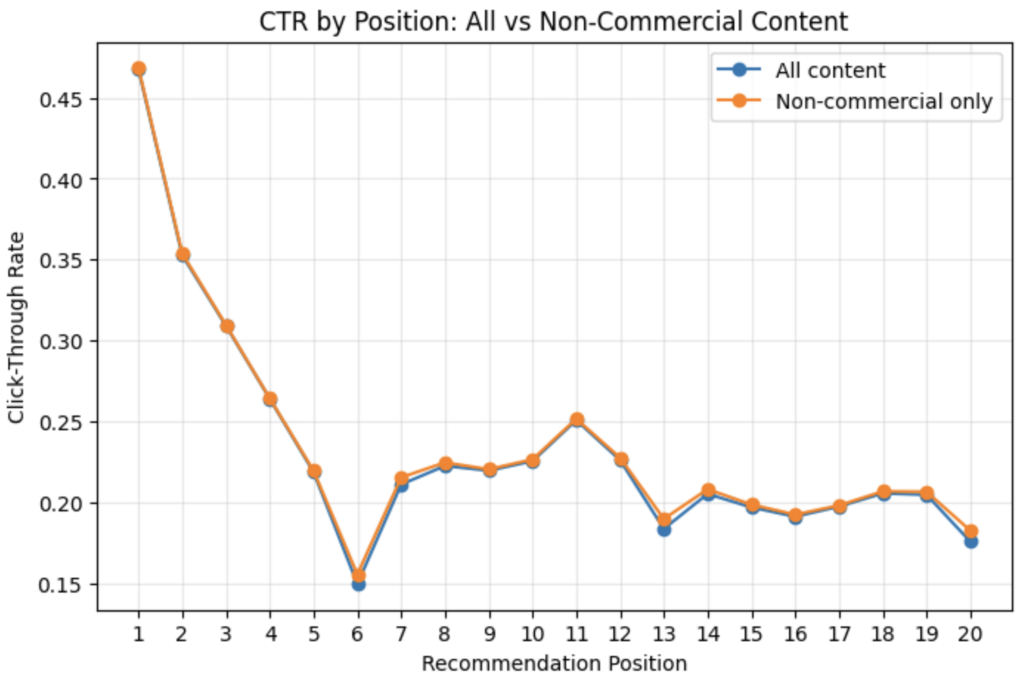
- **Block:** which group of seven positions an item belongs to
- **Within-block position:** the item's position within that seven-position group
- Positions **6, 13, 20, 27, …** all correspond to **within-block position 6**



- Block 1: 1, 2, 3, 4, 5, **6** , 7
- Block 2: 8, 9, 10, 11, 12, **13** , 14
- Block 3: 15, 16, 17, 18, 19, **20** , 21

Commercial Content Alone Cannot Explain the CTR Dip

- Removing commercial content did not eliminate the CTR dip.
- Therefore, commercial exposure alone is not sufficient to explain the recurring engagement penalty.



logistic regression

- **M1: Click~C(Block)+C(WithinBlockPosition)**
- Tests whether the sixth position within each seven-item block is associated with lower click probability.

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	Model	Sixth-slot Beta	Odds Ratio
0	M1: Position Structure	-0.280	0.756
1	M2: + Commercial	-0.237	0.789
2	M3: + Content Type/Format	-0.247	0.781
3	M4: + Topic & User Context	-0.242	0.785

If the added controls fully explained the anomaly:

βslot -> 0
ORslot -> 1

cross-fitted candidate-quality measure

- Revised approach
- 1. Predict expected click probability from position and context

$$\hat{p}_i = P(Click_i = 1 \mid Position, Content, Context)$$

- 2. Measure performance relative to expectation

$$Adjusted\ Performance_i = Click_i - \hat{p}_i$$

- 3. Use cross-fitting
- Each observation is evaluated by a model that was not trained on that observation.

within_block_position	n	avg_adjusted_quality
1	111366	0.007984
2	114617	0.008701
3	113772	0.009559
4	105832	0.009109
5	84946	0.004778
6	70512	0.001629
7	73323	0.004385

The sixth slot may not only suffer from a positional disadvantage; it also appears to receive candidates with weaker position-adjusted out-of-sample performance.

Conclusion and Implication

- **Conclusion**
- Commercial content partially explains the recurring sixth-slot CTR penalty.
- Observable content and user-context factors do not fully explain it.
- Sixth-slot candidates show weaker position-adjusted, out-of-sample performance than neighboring candidates.

- **Implication**
- The issue may involve the candidate-selection or serving mechanism, not only ad exposure.
- Product teams could audit these slots and use A/B tests to evaluate alternative candidate-selection strategies.

Reference

- Chen, J., Dong, Q., Li, H., He, X., Gao, Y., Cao, S., Wu, Y., Yang, P., Xu, C., Hu, Y., Ai, Q., & Liu, Y. (2025). Qilin: A multimodal information retrieval dataset with APP-level user sessions. In N. Ferro, M. Maistro, G. Pasi, O. Alonso, A. Trotman, & S. Verberne (Eds.), *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 3670–3680). Association for Computing Machinery. https://doi.org/10.1145/3726302.3730279